

QIXIA LUO

The University of British Columbia
Department of Mathematics
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EDUCATION

The University of British Columbia, PhD Applied Mathematics

September, 2025 - Present, Vancouver, BC, Canada

GPA: 93.5/100

- Advisor: Prof. Elina Robeva
- Research topics: Algebraic statistics; causal inference; data science; machine learning

Queen's University, MSc Mathematics

September, 2023 - July, 2025, Kingston, ON, Canada

GPA: 4.30/4.30

- Advisor: Prof. James A. Mingo
- Research topics: Random matrix theory; free probability
- Thesis: The north pole problem and recent developments in infinitesimal free probability

University of Toronto, HBSc Mathematics & Statistics

September, 2017 - June, 2022, Toronto, ON, Canada

GPA: 3.95/4.00

- Degree: Honours Bachelor of Science with High Distinction
- Program: Mathematics & Its Applications Specialist (Probability/Statistics)

RESEARCH INTERESTS

I am interested in the mathematical understanding of statistics, data science, and machine learning. In particular, I am interested in problems in algebraic statistics, causal discovery, and the algebraic geometry of neural networks.

MANUSCRIPTS IN PREPARATION

Structural Identifiability and Learning in Non-Gaussian Graphical Discrete Lyapunov Models, with Elina Robeva

CONFERENCES & WORKSHOPS

(Upcoming) Speaker, SIAM Conference on Computational Science and Engineering

February 22 - 26, 2027, Pittsburgh, PA, USA

- I have been invited to participate and give a 20 minutes talk in the session *Regression & Causality on Graphs*

(Upcoming) Speaker, Joint Mathematics Meeting (JMM)

January 12 -15, 2027, Chicago, IL, USA

- I have been invited to participate and give a 20 minutes talk in the session *Algebraic Statistics Across the Sciences: Models from Biology, Networks, and Learning*

(Upcoming) Speaker, Canadian Mathematics Society (CMS) Winter Meeting

December 11 -14, 2026, Montréal, QC, Canada

- I have been invited to participate and give a 20 minutes talk in the session *Applied and Computational Algebraic Geometry*

Attendant & Speaker, Women in Mathematics of Data Science Workshop

August 4 -14, 2026, Vancouver, BC, Canada

- I will work in Prof. Kathlén Kohn's group on neuroalgebraic geometry
- I gave a 20 minutes talk on structural identifiability and learning in graphical discrete Lyapunov models

Speaker, International Congress on Mathematical Software

July 20 - 23, 2026, Waterloo, ON, Canada

- I gave a 20 minutes talk titled *Graphical Discrete Lyapunov Models: Structural Identifiability and Learning* in the session *Biological Applications of Computer Algebra*
- I received a scholarship that covered my registration fees

Speaker, TENORS Workshop 2: Computational Methods for Tensors in Mathematics and Quantum Physics

July 13 - 17, 2026, Trento, TN, Italy

- I gave a 20 minutes talk titled *Graphical Discrete Lyapunov Models: Structural Identifiability and Learning*

Speaker, Society for Industrial and Applied Mathematics Annual Meeting 2026 (SIAM AN26)

July 6 - 10, 2026, Cleveland, OH, USA

- I gave a 20 minutes talk titled *Graphical Discrete Lyapunov Models: Structural Identifiability and Learning* in the session *Applications of Tensor Ranks and Decompositions*
- I served as the student representative of the UBC SIAM student chapter and received a travel stipend

Attendant, Upper Bound AI Conference

May 19 - 22, 2026, Edmonton, AB, Canada

- I received a 2026 Upper Bound Talent Bursary that covered all my travel, accommodation, food, and registration expenses

Attendant, UBC SIAM Student Chapter Applied Mathematics Meeting

October 25, 2025, Vancouver, BC, Canada

Attendant, Ontario Graduate Mathematics Conference (OGMC)

May 30 - June 1, 2025, Toronto, ON, Canada

Attendant, AI4GoodLab

May, 2020 - June, 2020, Edmonton, AB, Canada (remote due to pandemic)

- 7 weeks full-time workshop with a focus on the social implications of AI
- Amongst a small cohort of women and gender diverse individuals selected across Canada to participate

TEACHING & MENTORING

Summer Undergraduate Reading Program Mentor, The University of British Columbia

June - July, 2026, Vancouver, BC, Canada

- Mentored a group of five undergraduate students on a reading project titled *Finite Frames and its Applications* over two months. I structured and designed all learning materials, met with the students for two hours each week, and supervised final projects

Teaching Assistant, The University of British Columbia

September, 2025 - present, Vancouver, BC, Canada

- Graded homework assignments and midterms for Differential Calculus with Applications (MATH 100), Differential Calculus and Multivariable Calculus (MATH 253), Introduction to Discrete Mathematics (MATH 341)
- Tutored students in the Math Learning Center (MLC)
- Invigilated exams

Teaching Assistant, Queen's University

September, 2023 - August, 2025, Kingston, ON, Canada

- Graded homework and exams for linear algebra, integral and differential calculus, complex analysis, vector analysis
- Facilitated tutorials for integral and differential calculus
- Proctored exams

EMPLOYMENT

Intern, International Air Transport Association (IATA)

July, 2022 - April, 2023, Montréal, QC, Canada

- Conducted business data analytics and created reports
- Supported the global operations for IATA Training in a large international team setting

OTHER TALKS

Reading Group Speaker, The University of British Columbia

- Introduction to Free Probability and Random Matrices; *Learning Seminar on Tensors and Causal Inference*
- Representation Theory of $GL(\mathbb{C}^n)$; *Reading Group on Algorithms in Invariant Theory*

Departmental Seminar Speaker, Queen's University

- Stieltjes Transform and its Inversion; *Random Matrix Theory Seminar*
- Joint Infinitesimal Free Cumulants of Certain Unitarily Invariant Matrix Ensembles and their Transposes; *Random Matrix Theory Seminar*
- Multiplicative Linear Functionals on Unital Banach Algebras; *Operator Algebras Learning Seminar*

- Computing Maximal Ideal Spaces on Unital Banach Algebras; *Operator Algebras Learning Seminar*
- Commutative C^* -Algebras; *Operator Algebras Learning Seminar*
- The Johnson-Lindenstrauss Lemma; *Graduate Student Seminar*

GRADUATE AWARDS & SCHOLARSHIPS

- Four Year Doctoral Fellowship, *The University of British Columbia*, \$72,800 total over 4 years
- Tuition Award, *The University of British Columbia*, full tuition coverage over 4 years
- President's Academic Excellence Initiative PhD Award, \$840 per annum, *The University of British Columbia*
- The Senator Frank Carrel Fellowship, *Queen's University*, \$30,000 total over 2 years

UNDERGRADUATE AWARDS & SCHOLARSHIPS

- Howard Ferguson Scholarship, *University of Toronto*, \$12,000 total over 4 years
- The Coxeter Scholarship In Mathematics, *University of Toronto*, \$1480, one time recipient
- Galois Fund, *University of Toronto*, \$750 total, two times recipient
- C L Burton Open Scholarship, *University of Toronto*, \$400, one time recipient
- Dr. James A. & Connie P. Dickson Scholarship In Science & Mathematics, *University of Toronto*, \$400, one time recipient
- Dean's Scholar, *University of Toronto*, 2018 Summer, 2020 - 2022 Winters
- Hugh G. White Memorial Scholarship, *Actuarial Foundation of Canada*, \$1200, one time recipient

VOLUNTEERING & SERVICES

- Application evaluation committee for the 2026 AI4GoodLab cohort
- Grader for the 2025 Canadian Open Mathematics Contest (COMC)
- Treasurer of the UBC SIAM Student Chapter

COMPUTING SKILLS

Familiar with Python, R, JavaScript, MATLAB, Julia, Macaulay2

MISCELLANEOUS

- Citizenship: Canada
- First name pronunciation: /tʃiːʃə/ or *chee-shuh*
- This *curriculum vitae* is up-to-date as of *September 7, 2026*